CSC429: Lab #1

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1. Open-Source Data Resources

Resource 1: MNIST

Link: https://www.kaggle.com/datasets/hojjatk/mnist-dataset
Description: The MNIST dataset contains 70,000 handwriting digit samples. Frequently used to evaluate Federated Learning models for image classification.

Resource 2: CIFAR-10

Link: https://www.cs.toronto.edu/ kriz/cifar.html
Description: The CIFAR-10 dataset contains 60,000 image samples with 10 classes. Often used to evaluate Convolutional Neural Networks.

2. Open-Source ML Algorithms

Classification: Convolutional Neural Network (CNN)

Link: Keras Sequential Model
Description: A Convolutional Neural Network (CNN) is a deep learning algorithm that takes input features and passes them through convolutional and pooling operations to detect patterns. Very commonly used for image classification.

Regression: Linear Regression

Link: Scikit-Learn Linear Regression
Description: Linear regression is a foundational ML algorithm used to model outcomes based on one or more independent variables using basically a line. Linear regression algorithms are extremely inexpensive and often able to be trained locally for Federated Learning applications.

Clustering: K Means Clustering

Link: Scikit-Learn K-Means
Description: BriK-Means is an unsupervised ML algorithm that partitions unlabelled data into clusters based on feature similarity. Scikit-Learn has an open-source library that is very handy for clustering tasks.

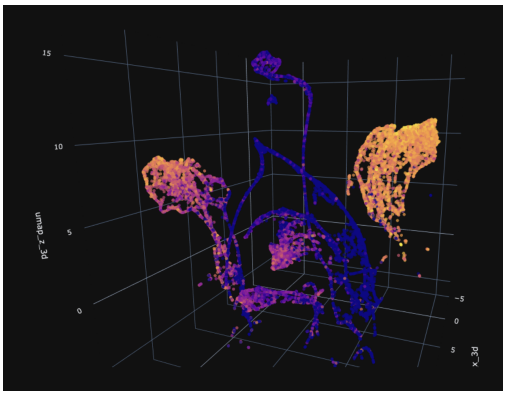

Figure 1: K Means Clustering Example 3d representation

3. Security and Privacy Applications

Application 1: Federated learning in IoT devices.

Federated Learning (FL) allows IoT devices to collaboratively train machine learning models using local data, such as vehicle sensor data or user network traffic, without sending raw data to a central server. ML is very useful here because it enables distributed networks to learn from vast amounts of real-world data while strictly preserving user privacy, as only the computed model updates are shared rather than the sensitive data itself.

Application 2: Privacy-preserving intrusion detection in IoT.

For industrial IoT environments, it is important to have robust Intrusion Detection Systems. However, oftentimes industrial network traffic is sensitive, and transferring it to a central server risks exposing proprietary operational data. Machine Learning is useful here because models, such as a 1D CNN, can be deployed to monitor and classify malicious traffic (like a DoS) locally.

Application 3:

Describe the application briefly and explain why ML is useful for it.

References (Everything Looked at)

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